

Tech & Teach

Digital Chip Design Program

Design RTL

names:

Shaher Alharbi

Suliman w Alrumaikani

Abdullah AlJumah

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Task :0.1

2026\06\29

Task topic:

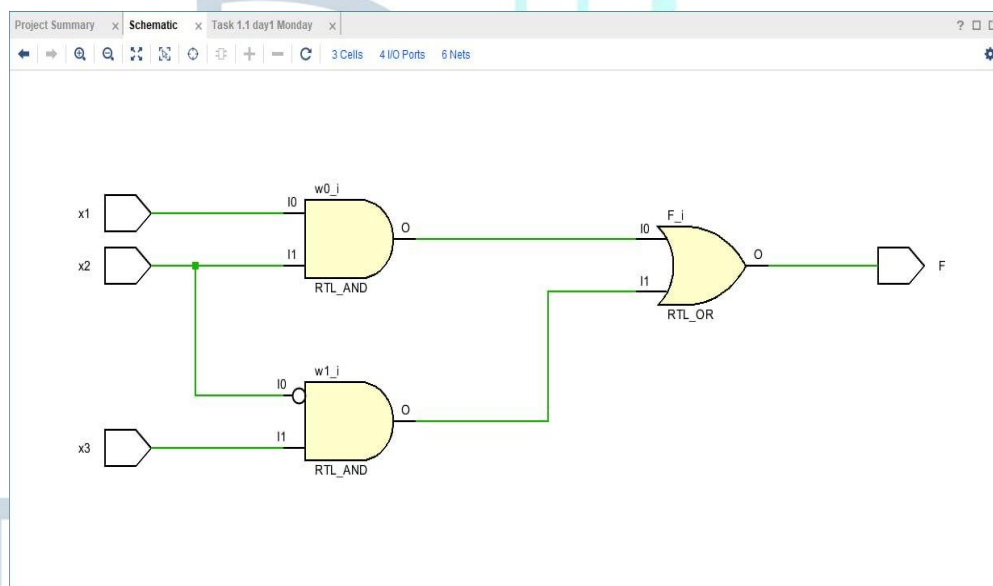
Gate-level

Version 1.0

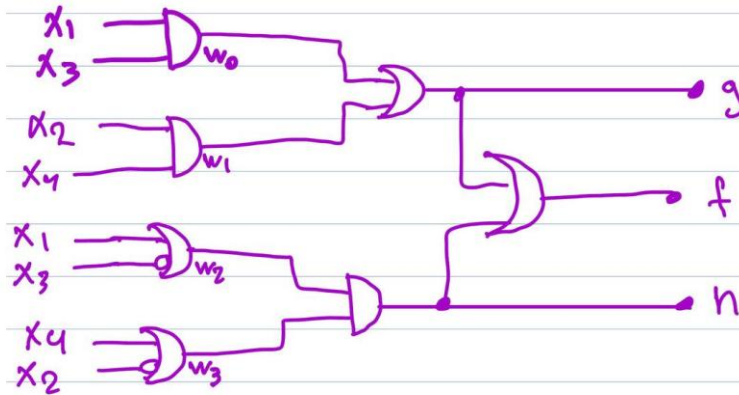
Task: 1.1

```
Project Summary x Schematic x Task 1.1 day1 Monday x ? □ □
C:/Users/HUAWEI/Desktop/Vivado/Task 1.1 - Day 1 Monday/Task 1.1 - Day 1 Monday/srcs/sources_1/new/Task 1.1 day1 Monday

1 module Task (input logic x1, input x2, input x3,
2 output logic F);
3
4 logic w0, w1;
5 logic nx;
6 not ( nx , x2 );
7 and ( w0 , x1 , x2 );
8 and ( w1 , nx , x3 );
9 or ( F , w0 , w1 );
10 endmodule
```



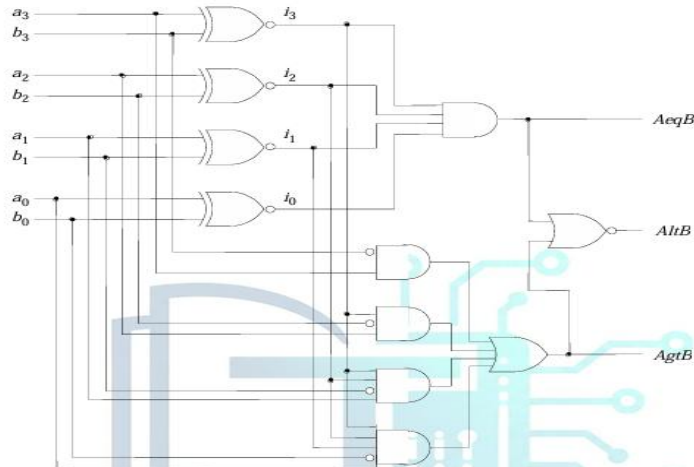
Task: 1.2



```
1 module comb_circuit(f, g ,h, x);
2
3 input [4:1]x;
4 output f, g, h;
5
6 wire [3:0]w;
7
8 and (w[0], x[1], x[3]);
9 and (w[1], x[2], x[4]);
10 or (w[2], x[1], ~x[3]);
11 or (w[3], ~x[2], x[4]);
12 or (g, w[0], w[1]);
13 and (h, w[2], w[3]);
14 or (f, g, h);
15
16 endmodule
```

Task: 1.3

Using SystemVerilog, implement the following logic circuit in gate level modeling



```
1 module task3 (  
2     input a3,input b3,input a2,input b2,input a1,input b1,input a0,input b0,  
3     output AeqB,output AltB,output AgtB);  
4     logic w0 , w1 , w2 ,w3;  
5     logic nb3 , nb2 , nb1 , nb0;  
6     logic w4 , w5 , w6 , w7;  
7     //the first layer  
8     xnor (w0,a3,b3);  
9     xnor (w1,a2,b2);  
10    xnor (w2,a1,b1);  
11    xnor (w3,a0,b0);  
12    //the second layer upper part(output from past input here)  
13    and (AeqB,w0,w1,w2,w3);  
14    //lower part in second layer  
15    not (nb3,b3);  
16    not (nb2,b2);  
17    not (nb1,b1);  
18    not (nb0,b0);  
19  
20    and (w4,nb3,a3);  
21    and (w5,w0,nb2,a2);  
22    and (w6,w0,w1,nb1,a1);  
23    and (w7,w0,w1,w2,nb0,a0);  
24  
25    or (AgtB,w4,w5,w6,w7);  
26    //last layer  
27    nor (AltB,AeqB,AgtB);  
28 endmodule
```

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Work distribution

Section	shaher	Suliman	Abdullah
Report	✓		
Task 1.1			✓
Task 1.2	✓		
Task 1.3		✓	

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